

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
#include <stdio.h>

#include <ctype.h>

#include <string.h>

const char *keywords[] = {
    "int", "float", "if", "else", "while", "for", "return", "char", "double", "void"
};

int numKeywords = 10;

// Check if string is a keyword
int isKeyword(char *str) {
    for (int i = 0; i < numKeywords; i++) {
        if (strcmp(str, keywords[i]) == 0)
            return 1;
    }
    return 0;
}

// Check if character is an operator
int isOperator(char ch) {
    char operators[] = "+-*/=%><!";
    for (int i = 0; operators[i] != '\0'; i++) {
        if (ch == operators[i])
            return 1;
    }
    return 0;
}

// Check if character is a separator
int isSeparator(char ch) {
```

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
char separators[] = "(){}";  
for (int i = 0; separators[i] != '\0'; i++) {  
    if (ch == separators[i])  
        return 1;  
}  
return 0;  
}
```

```
int main() {  
    FILE *file = fopen("input.txt", "r");  
    if (file == NULL) {  
        printf("Error: Could not open file.\n");  
        return 1;  
    }
```

```
    char ch;  
    int tokenCount = 0;
```

```
    while ((ch = fgetc(file)) != EOF) {  
        // Skip whitespace  
        if (isspace(ch)) continue;
```

```
        // Identifiers or Keywords (start with letter or underscore)
```

```
        if (isalpha(ch) || ch == '_') {  
            char word[50];  
            int i = 0;
```

```
            while (isalnum(ch) || ch == '_') {  
                word[i++] = ch;  
                ch = fgetc(file);
```

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
}  
  
word[i] = '\0';  
  
ungetc(ch, file); // put back the non-identifier character  
  
if (isKeyword(word))  
    printf("Keyword: %s\n", word);  
else  
    printf("Identifier: %s\n", word);  
  
tokenCount++;  
}  
  
// Numbers  
else if (isdigit(ch)) {  
    printf("Number: ");  
    while (isdigit(ch) || ch == '.') {  
        printf("%c", ch);  
        ch = fgetc(file);  
    }  
    printf("\n");  
    ungetc(ch, file);  
    tokenCount++;  
}  
  
// String literals  
else if (ch == '"') {  
    printf("String: \");  
    ch = fgetc(file);  
    while (ch != '"' && ch != EOF) {  
        printf("%c", ch);
```

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
        ch = fgetc(file);
    }
    printf("\n\n");
    tokenCount++;
}

// Character literals
else if (ch == '\\') {
    printf("Character: ");
    ch = fgetc(file);
    while (ch != '\\' && ch != EOF) {
        printf("%c", ch);
        ch = fgetc(file);
    }
    printf("\n\n");
    tokenCount++;
}

// Operators
else if (isOperator(ch)) {
    printf("Operator: %c\n", ch);
    tokenCount++;
}

// Separators
else if (isSeparator(ch)) {
    printf("Separator: %c\n", ch);
    tokenCount++;
}
```

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
// Unknown characters
else {
    printf("Unknown: %c\n", ch);
    tokenCount++;
}
}

printf("\nTotal Tokens: %d\n", tokenCount);

fclose(file);
return 0;
}
```

Text File:

```
int a = 10;
float b = 20.5;
char c = 'x';
string = "AAA"
```

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

OUTPUT:

```
Keyword: int
Identifier: a
Operator: =
Number: 10
Separator: ;
Keyword: float
Identifier: b
Operator: =
Number: 20.5
Separator: ;
Keyword: char
Identifier: c
Operator: =
Character: 'x'
Separator: ;
Identifier: string
Operator: =
String: "AAA"

Total Tokens: 18
```